

A Research Principle for the AI Era

Jianmin Liao

September 28, 2026

Research in the AI era should pursue intrinsic value while building the external support needed to sustain that pursuit.

Under the assumptions below, maximizing expected intrinsic value requires an initial phase devoted entirely to external return. An optimal strategy then mixes intrinsic and external work. With enough time, its expected intrinsic allocation share is nondecreasing, even when returns vary randomly across cycles.

Intrinsic value means the scientific understanding or other substantive contribution that the researcher ultimately values. External return means support convertible into research funding, measured in the same units as expenditure. The model treats external support as a means of sustaining research.

Assumptions

Consider n research cycles. In cycle i , let c_i be expenditure and $p_i \in [0, 1]$ the fraction allocated to intrinsic work. Normalize the initial expenditure to $c_1 = 1$.

1. Linear returns with iid random coefficients. Intrinsic value, external return, and the next research budget satisfy

$$I_i = a_i p_i c_i, \quad E_i = b_i (1 - p_i) c_i, \quad c_{i+1} = d_i E_i.$$

The vectors (a_i, b_i, d_i) are independent and identically distributed across cycles. Their components may be dependent within a cycle. All coefficients have finite upper bounds, and a_i has a strictly positive lower bound. Write

$$b = \text{ess inf } b_i, \quad d_i \geq 1 \text{ almost surely}, \quad \mathbb{E}[d_i] > 1.$$

Thus b is the greatest almost-sure lower bound on the external-return coefficient. The funding conditions mean that external support is fully carried into the next budget and has a positive probability of expanding it. The common distribution stays fixed across cycles, although realized coefficients can change.

2. A positive and growing safety balance. Let the cumulative funding surplus satisfy

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

Require, almost surely in every cycle,

$$B_i \geq \max\{\delta, \rho B_{i-1}\}, \quad 1 < \rho < b, \quad 0 < \delta \leq b - 1.$$

The floor δ establishes a safety reserve, and ρ requires it to grow. The coefficient bound makes the initial reserve feasible and leaves room for intrinsic work. The initial expenditure is supplied as seed funding; balances are assessed at cycle boundaries.

3. Decisions before current returns are known. The researcher observes the current budget and balance and chooses p_i before observing the current coefficients. Past coefficients need not be observed. The common coefficient distribution is known, and the sole objective is

$$\max \mathbb{E} \left[\sum_{i=1}^n I_i \right].$$

Each cycle's intrinsic value receives equal weight. External return matters through subsequent funding and the safety constraints. This information structure separates observable resources from uncertain research productivity.

The optimal allocation

Theorem 1. *Under these assumptions, there are finite integers $1 \leq K < N$, depending on the common distribution and the model parameters, with the following property. For every horizon $n \geq N$, every optimal strategy devotes its first $n - K$ cycles entirely to external return:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

There is an optimal strategy that then uses the minimum external share needed to guarantee the safety balance:

$$p_i = 1 - \frac{c_i + (\rho - 1)B_{i-1}}{bc_i} \in (0, 1) \quad (n - K < i \leq n).$$

For this strategy, the expected intrinsic allocation share is nondecreasing:

$$\begin{aligned} 0 &= \mathbb{E}[p_1] = \cdots = \mathbb{E}[p_{n-K}], \\ 0 &< \mathbb{E}[p_{n-K+1}] \leq \cdots \leq \mathbb{E}[p_n]. \end{aligned}$$

Equivalently, the expected external share $1 - \mathbb{E}[p_i]$ is nonincreasing. The strategy requires only the current budget, balance, cycle number, and the known distribution.

The phase structure already holds whenever $n > K$. The possibly larger threshold N ensures monotonicity of the expected proportions. These expectations average over random histories; individual trajectories can fluctuate. Favorable external returns can leave the realized balance above its required minimum.

A numerical example

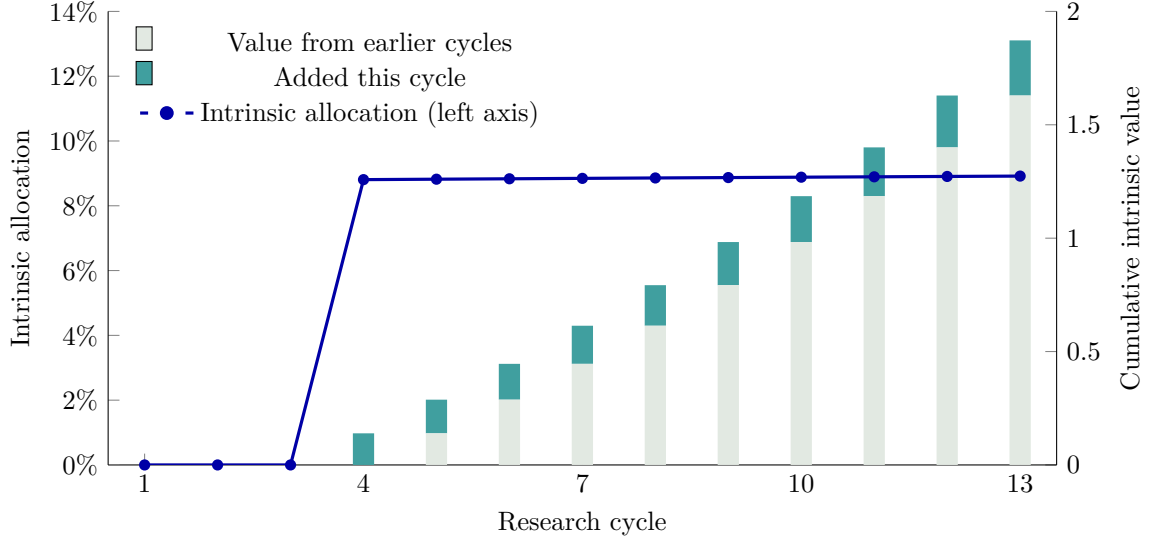


Figure 1: An optimal trajectory with the same fixed linear functions in every cycle: three cycles of full external focus, followed by ten cycles of mixed allocation. The line shows the intrinsic allocation share (left axis). Each full bar shows cumulative intrinsic value (right axis); its darker segment is the intrinsic value added in that cycle, and its lighter segment is the value accumulated in earlier cycles.

For a single fixed-coefficient illustration, use

$$a_i = 1, \quad b_i = 1.11, \quad d_i = 1.05,$$

with $n = 13$, $\rho = 1.05$, and $\delta = 0.1$. Constant coefficients are a degenerate iid distribution and satisfy $\mathbb{E}[d_i] > 1$. The Bellman recursion below gives $K = 10$: three cycles of full external focus, followed by ten cycles of mixed allocation.

Here $d_i = \rho$, so during the mixed phase the reserve-to-budget ratio satisfies

$$\frac{B_i}{c_{i+1}} = \frac{B_{i-1}/c_i}{1 + (\rho - 1)B_{i-1}/c_i} < \frac{B_{i-1}}{c_i}.$$

Consequently, the intrinsic share increases throughout this phase. For these fixed coefficients, monotonicity already holds for every $n > K$.

The intrinsic share rises from 8.81% in cycle 4 to 8.92% in cycle 13. The corresponding intrinsic increments are 0.1394 and 0.2431, and cumulative intrinsic value reaches 1.8722. This is one deterministic trajectory with no averaging. The same figure combines the allocation line with cumulative-value bars whose darker portions show the current increments.

Proof

1. Feasible actions and the Bellman recursion. After the first cycle, the states satisfy $\delta \leq B \leq c$. Indeed, if $B \leq c$, then

$$B' = B + E - c \leq E \leq d_i E = c'.$$

The safety constraint preserves the lower bound. Write $u = (1 - p)c$ for the expenditure allocated to external work. At a state (c, B) , its feasible interval is exactly

$$\frac{c + (\rho - 1)B}{b} \leq u \leq c.$$

This follows from the essential lower bound b and the almost-sure safety requirement. The interval has positive length because $B \leq c$ and $b > \rho$. In the first cycle, the interval is $[(1 + \delta)/b, 1]$.

The current coefficient a_i is independent of the information used to choose p_i . Dividing the objective by the positive constant $\mathbb{E}[a_i]$ therefore gives expected stage reward $pc = c - u$. Set

$$w = \mathbb{E}[b_i], \quad v = \mathbb{E}[b_i d_i], \quad v \geq w \geq b > \rho > 1.$$

Induction shows that the optimal normalized expected value with m cycles remaining has the form

$$V_m(c, B) = A_m c + D_m B, \quad A_0 = D_0 = 0.$$

Define $S_m = vA_m + wD_m$. Given the continuation coefficients, the current objective is

$$(1 - D_{m-1})c + D_{m-1}B + (S_{m-1} - 1)u.$$

Thus the upper endpoint is uniquely optimal when $S_{m-1} > 1$. When $S_{m-1} \leq 1$, choose the lower endpoint, including at a tie. The lower choice gives

$$\begin{aligned} A_m &= 1 - D_{m-1} + \frac{S_{m-1} - 1}{b}, \\ D_m &= D_{m-1} + \frac{(\rho - 1)(S_{m-1} - 1)}{b}. \end{aligned}$$

The upper choice gives

$$A_m = vA_{m-1} + (w - 1)D_{m-1}, \quad D_m = D_{m-1}.$$

These updates establish the claimed linear value function.

2. A single switch between phases. While the lower endpoint is chosen, $D_m \leq 0$. Direct substitution gives

$$\begin{aligned} S_m &= \frac{v + w(\rho - 1)}{b} S_{m-1} + \frac{v(b - 1) - w(\rho - 1)}{b} \\ &\quad + (w - v)D_{m-1} \geq \rho S_{m-1} + b - \rho. \end{aligned}$$

Starting from $S_0 = 0$, this sequence increases until it crosses 1 in finite time. Let

$$K = \min\{m \geq 1 : S_m > 1\}.$$

After the crossing, the upper update gives

$$S_m = vS_{m-1} + (w - v)D_{m-1} > S_{m-1} > 1.$$

The coefficient D_m then remains fixed and nonpositive. Therefore every decision with more than K cycles remaining uniquely chooses $u = c$, and every later decision admits the lower-endpoint optimum. The first-cycle interval has a different lower endpoint, but the same strictly positive decision coefficient when $n > K$. This proves the phase structure and the displayed allocation formula.

3. Monotonicity of the expected proportions. Write $r = B/c \in [0, 1]$. In the mixed phase,

$$1 - p = \frac{1 + (\rho - 1)r}{b}.$$

It therefore suffices to prove that the expected reserve-to-budget ratio decreases during that phase, once the preceding full-external phase is long enough.

For a fresh draw (b_i, d_i) , the two state transitions are

$$\begin{aligned} F_i(r) &= \frac{r + b_i - 1}{b_i d_i} && \text{under full external allocation,} \\ G_i(r) &= \frac{1}{d_i} \left(1 - \frac{b}{b_i} \frac{1 - r}{1 + (\rho - 1)r} \right) && \text{under mixed allocation.} \end{aligned}$$

Both maps preserve $[0, 1]$ and are strictly increasing. Each F_i contracts distances by at most $1/b < 1$. Iterating the iid maps from an independent infinite past gives a stationary distribution π . Coupling any other initial state to it using the same draws makes their distance at most b^{-t} after t steps.

If R has distribution π and is independent of a fresh coefficient draw, stationarity gives

$$\mathbb{E}[1 - R] = \frac{\mathbb{E}[1 - 1/d_i]}{1 - \mathbb{E}[1/(b_i d_i)]} > 0.$$

The numerator is positive because $d_i \geq 1$ and $\mathbb{E}[d_i] > 1$. Moreover,

$$F_i(r) - G_i(r) = \frac{1 - r}{b_i d_i} \left(\frac{b}{1 + (\rho - 1)r} - 1 \right) > 0 \quad (r < 1).$$

Consequently, applying G_i to a stationary starting state produces a distribution that can be coupled below π , with a strictly smaller mean. Applying the same subsequent draws preserves strict order, because every G_i is strictly increasing. Thus successive mixed-phase distributions started from π have strictly decreasing means.

There are only K mixed cycles. As the full-external prefix length $n - K$ grows, its final state distribution converges to π . The maps G_i have Lipschitz constants at most ρ , so each of these finitely many mixed-phase expectations converges to its stationary-start counterpart. All adjacent limiting differences are strictly positive. They therefore remain positive for every sufficiently long prefix. Choose a finite $N > K$ beyond this point. If $K = 1$, there are no mixed-phase comparisons and $N = K + 1$ suffices.

At the phase switch, the external proportion drops from 1 to at most $\rho/b < 1$. Combining this drop with the mixed-phase comparisons proves the expected monotonicity for every $n \geq N$. Finally, the constructed optimal policy uses only the cycle number and the observed pair (c_i, B_{i-1}) . It remains admissible and optimal when past coefficients are unobserved. $\square$

Even when intrinsic value is the sole objective, building external support first can be the optimal way to sustain an increasing intrinsic allocation over time in expectation.